

----- Active space -----

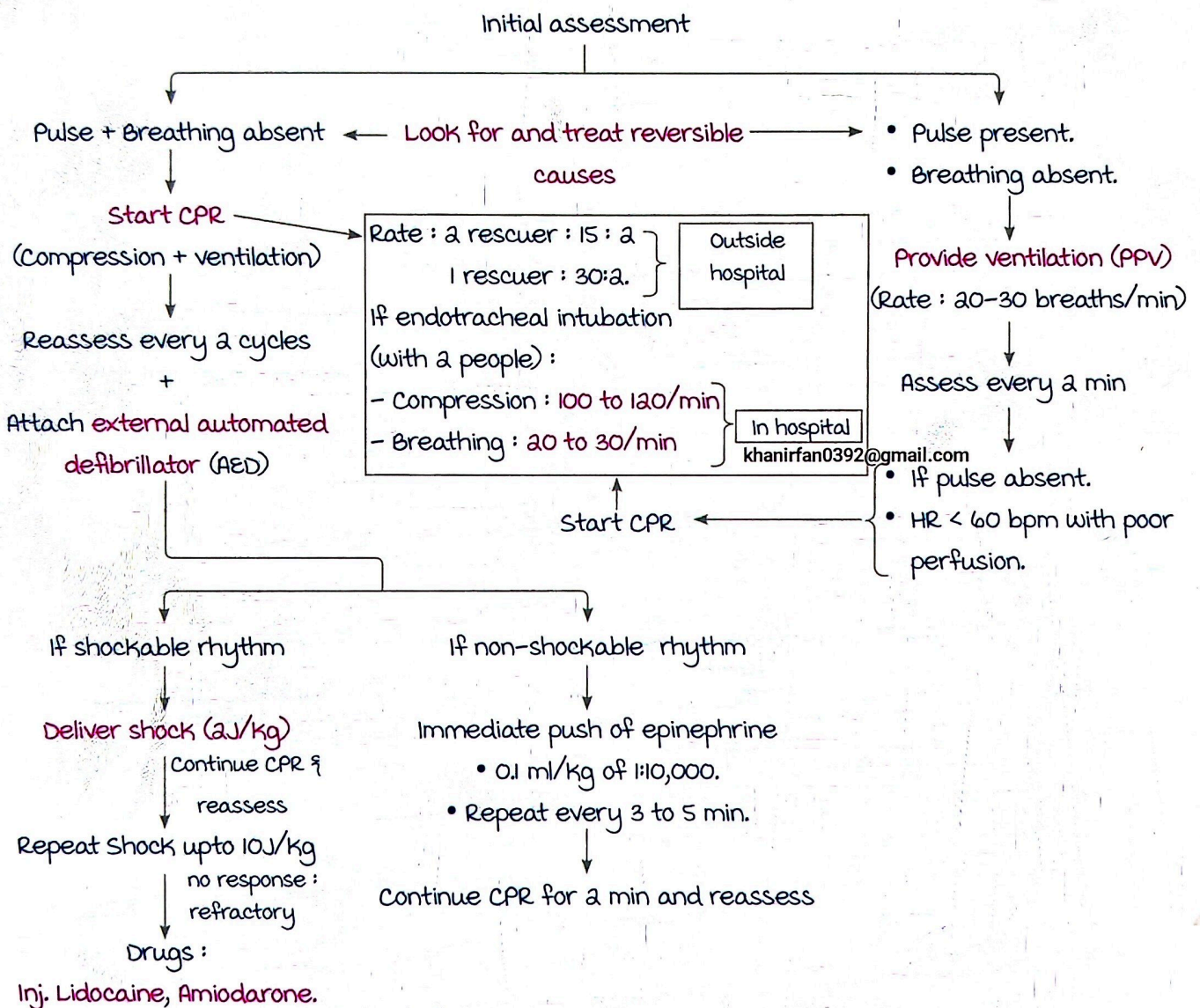
PAEDIATRIC RESUSCITATION

Causes of cardiopulmonary arrest in a child :

- Respiratory causes > cardiac causes.
 - Respiratory failure.
 - Terminal stages of shock.

Cardiopulmonary Resuscitation (CPR)

00:01:08



INITIAL ASSESSMENT

----- Active space -----

Verify scene safety

- Position : Supine on a flat, hard surface.
- Immobilise patient if fracture suspected (RTA).

Check for responsiveness : Look for any body movements

↓ if unresponsive

Shout for help & activate an emergency response system (if in public)

+

Check for :

- Breathing chest movements.
- Pulse : Brachial artery (infant), carotid or femoral (> 1 yr).

Simultaneously
checked
for < 10 sec

CIRCULATION :

Chest compressions to be performed.

Site : Lower 1/3rd of sternum.

Rate : 100 to 120 min.

Depth : 1/3rd of Antero-posterior (AP) diameter of chest
 (Always allow full recoil to ensure filling of heart)

Push hard
Push fast

AIRWAY :

1. Head tilt chin lift.
2. Jaw thrust.

BREATHING :

Non-invasive :

Bag and -mask ventilation.

Invasive : Endotracheal intubation

- Preferred in :
 - In hospital CPR
 - Prolonged CPR

Endotracheal intubation :

Type of tube : Cuffed

Size of ET tube = $\frac{\text{Age (in yr)} + 4}{4}$ Cuff pressure : < 20 to 25 cm of H₂O
(to prevent pressure necrosis)

Depth of insertion : 3 x Age (in yrs)

Cricoid pressure : Routine use to be avoided (To prevent cartilage damage).

Rate of breathing : 20 to 30 breaths/min (1 breath every 2-3 sec).

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Miscellaneous

----- Active space -----

VASCULAR ACCESS :

Goal : Administration of medications.

Preferred route : IV route.

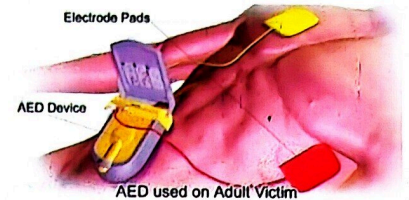
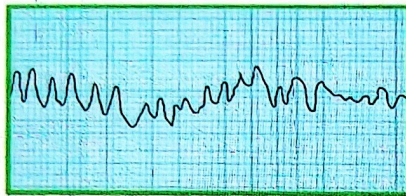
- Large accessible vein : Femoral vein.

Alternate route : Intraosseous route (I.O)

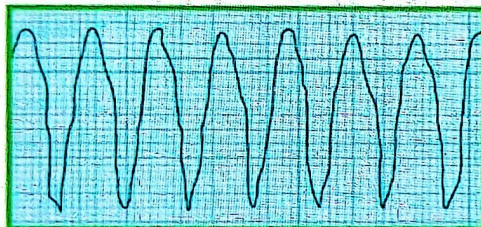
- Proximal tibia > Distal femur (Below tibial tuberosity).

AUTOMATED EXTERNAL DEFIBRILLATOR :

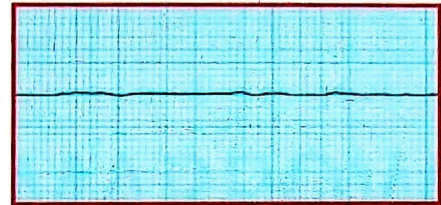
- Displays rhythm to decide management.
- Causes asynchronous depolarization of myocardium (Shock).

**1. Shockable rhythm (Wide QRS) :**

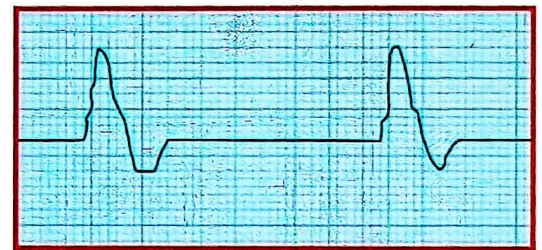
- a. Irregular rhythm/QRS
ventricular fibrillation (V-fib)



- b. Regular rhythm/QRS
Pulseless ventricular tachycardia (PVT)

2. Non-shockable rhythm :

- a. Asystole



- b. Pulseless electrical activity (PEA)

REVERSIBLE CAUSES :

6H and 4T.

- Hypovolemia
- Hypothermia
- Hypoglycemia
- Hydrogen ion (Acidosis).
- Hypo/hyperkalemia

Thromboembolism. — Pulmonary
Toxin. — Coronary
Tamponade (Pericardial).
Tension pneumothorax.